



Capstone Project 2

CMU-SE 451

Project Proposal

Version 1.0

Date: 20 - May - 2024

DnD Translate - Sign Language

Submitted by

Duy, Huynh Gia

Minh, Huynh Tan Quang

Sy, Nguyen Van

Dinh, Le Nguyen Nhut

Approved by

Proposal Review Panel Representative:

Name

Signature

Date

Capstone Project 2 - Mentor:

Name

Signature

Date

Thuan, Nguyen Tan Ph.D.

Table of Contents

1. Project Overview.....	3
2. Executive Summary.....	4
3. Project Background.....	4
4. Project Objectives.....	6
5. Scope and Requirements.....	6
5.1. Scope.....	6
5.2. Requirements.....	6
5.3. System context.....	7
6. Methodology and Approach.....	8
6.1. Technical Constraints.....	8
6.2. Business constraints.....	9
7. Project Timeline.....	9
8. Resource Allocation.....	9
9. Budget and Cost Estimate.....	10
10. Risk Management.....	11
11. Stakeholder Engagement.....	12
12. Project Constraints.....	13
13. References.....	15
14. Attachment: DESCRIPTION OF PRODUCT REQUIREMENTS FORM.....	15
15. Approval.....	15

REVISION HISTORY

Version	Date	Description of Changes	Author
1.0	15/02/2024	Initial document	Team

1. Project Overview

PROJECT INFORMATION

Project acronym	SLDS		
Project Title	Sign Language Detection System		
Start Date	15 - Feb - 2024	End Date	20 - May - 2024
Lead Institution	International School, Duy Tan University		
Project Mentor	Name: Thuan, Nguyen Tan Ph.D. Email: thuannt.it.dtu@gmail.com Tel: 0905.626.276		
Project Manager & contact details	Name: Duy, Huynh Gia Email: huynhduy25072002@gmail.com Tel: 0762.555.759		
Partner Organization			
Project Web URL			
Team members	Name	Email	Tel
25211216743	Sy, Nguyen Van	vansy01682937260@gmail.com	0377.445.085
25211208352	Minh, Huynh Tan Quang	liverpool990minh12@gmail.com	0929.097.985
25217211295	Dinh, Le Nguyen Nhut	lenguyennhutdinh@gmail.com	0965.925.991

2. Executive Summary

Our system is not just a means of communication, but also a transformative tool, opening the doors to connection and understanding between individuals, regardless of their form of communication. Through our website, users can participate in remote video calls and experience a diverse communication space.

Easy to use and access, our website connects users through video calls where they can communicate in sign or spoken language. Through the use of real-time translation technology, gestures and sign language are automatically converted into spoken or written language, facilitating the exchange of ideas and information.

Similar to Google Translate, our website acts as a flexible communication companion, focusing on converting visual input. By simply pointing the camera at the speaker or signer, our technology automatically interprets gestures and converts them into the desired form of communication, including spoken, written, or spoken language. sign language.

3. Project Background

In the age of technology, connecting deaf and mute people with the world through video calls is necessary. This has spurred extensive research and development in the field of sign language interpretation systems. Here, we explore existing solutions that have paved the way for our own system.

Example:

- <https://www.ngonngukyhieu.com/>
- <https://viesign.org.vn/>

The corresponding website name	General assessment
viesign	Comprehensive: This is the first and only MOOC (Massive Open Online Course) platform to learn Sign Language in Vietnam. Up-to-date: The website is regularly updated with the latest news and research on sign language.

ngonngukyhieu	Comprehensive: This website shares free information about sign language for the deaf and everyone. Up-to-date: The website is regularly updated with the latest news and research on sign language.
---------------	---

For our project.

The common features that we can provide	The limitations
- The system allows users to translate sign language into regular language and vice versa - The system uses AI to analyze images and make predictions about sign language. - The system allows users to connect with each other via video call, automatically translating languages right on the call	- Understanding: Current AI systems may struggle to fully understand the context and meaning of sign language gestures due to their complexity and variety. - Diversity of sign languages: There are more than 200 different sign languages in the world and each language has its own distinct expressions. - Limited data: The lack of large and semantically rich sign language data limits the development of accurate translators. - Video call connection: While there are platforms that support real-time sign language translation via video call, they typically only support a limited number of simple words and phrases, and require Larger and higher quality databases to improve translation models.

To create an exceptional website that can compete with others, we have incorporated unique and innovative features such as:

The system provides comprehensive solutions for connecting deaf and mute people with everyone. It can recognize sign languages, analyze them with AI, and convert them into natural language in the most natural way.

4. Project Objectives

- Facilitate Communication: Build a platform that allows deaf and mute people to communicate naturally and effectively through video calls.
- Sign Language Translation: Develop automatic sign language translation technology that uses machine learning to translate sign language in real time.
- Integrating Machine Learning Technology: Uses advanced machine learning models to accurately recognize and translate sign language.
- Ensuring Security: Protects users' personal information and communication data.
- Multi-Language Support: Provides services for many different sign languages, not limited to a specific language.
- Community Development: Create conditions for the deaf and mute community to develop and support each other.
- Research and Development: Continue to research and update technology to improve service quality.

5. Scope and Requirements

5.1. Scope

This document provides an overview of how the project will be developed. It includes the description of business needs, the proposed solution, the financial forecast, and some constraints that are involved in the project.

The document provides a comprehensive master plan for each phase of software development based on the processes that have been selected. This document is made for senior management to put forward a proposal.

5.2. Requirements

Functional Requirements:

- **User Authentication:** Implement user registration and login to manage user profiles.
- **Data Management:** Store and retrieve videos, user information, and call history.

- **Integration with External Systems:** Link to cloud storage services and third-party APIs.
- **Sign Language Translation:** Use machine learning to translate sign language in real time.

Non-functional Requirements:

- **Performance:** Ensure fast response times and smooth video processing.
- **Reliability:** Maintain stable operation with low error rates.
- **Security:** Protect user data and call information.
- **Usability:** Provide a user-friendly interface for easy access.
- **Scalability:** Support more users without compromising performance.
- **Maintenance:** Allow for easy updates and system maintenance.

5.3. System context

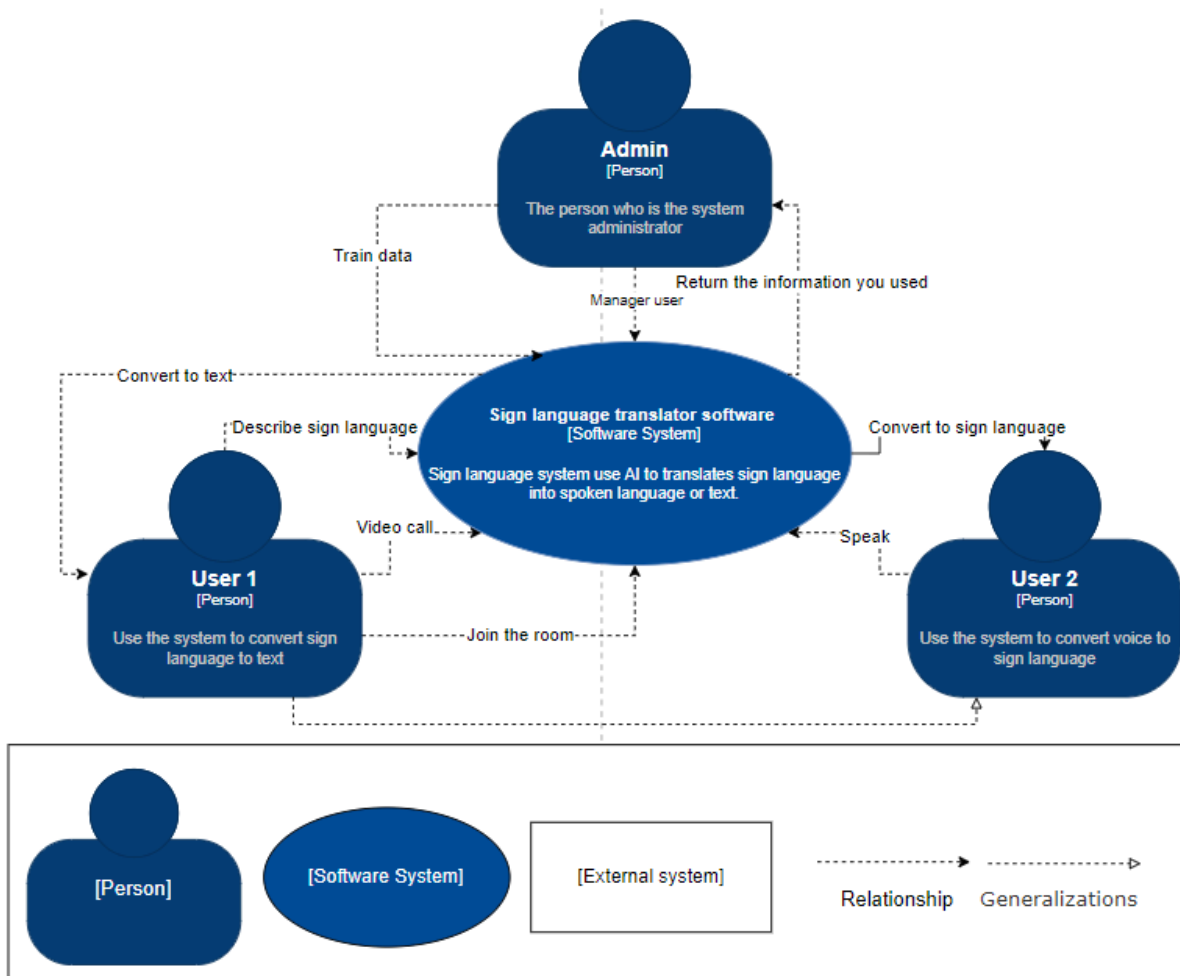


Figure 1: System context overview**System context description****Users:**

- Join the room
- Video call
- Describe language

Sign language system:

- Convert language
- Learning data use provide

Admin:

- Pass data for training
- Manage users

6. Methodology and Approach

Our team is building a "Sign language detection system" - a diagnostic tool to support the disability community. By leveraging the power of artificial intelligence (AI) and image processing technology, our system analyzes images to help users translate more accurately.

We will utilize the following technologies to create this website:

- TensorFlow: Deep learning AI model, providing accurate predictions.
- Google Sign-In API and Facebook Graph API are used to support registration or login via email or Facebook.

Communication and collaboration will be facilitated through regular project meetings, daily stand-ups, and online collaboration tools such as Zalo or Google Meet. We will maintain open channels of communication with stakeholders, providing regular updates on project progress, milestones, and any changes or issues that arise.

6.1. Technical Constraints**Server:**

- Programming Language: Python.
- Database: PostgreSQL

- Framework: fastapi.
- Libraries: Tensorflow, websockets.
- Operating System: Windows, Linux, macOS.

Client:

- Programming language: JavaScript.
- Framework: NextJS.
- Libraries: ReactJS.
- Operating System: Windows, Linux, macOS.
- Web Browser: Chrome, Firefox, Microsoft Edge, ...

6.2. Business constraints

- Resource: 4 people.
- Budget: \$5000.
- Time: The project must be completed within 4 months.

7. Project Timeline

No	Task Name	Duration	Start	Finish
1	Start Up	31 days	22/01/24	21/02/24
1.1	Project Kick Off Meeting	1 day	22/01/24	22/01/24
1.2	Discuss project ideal	10 days	23/01/24	01/02/24
1.3	Create a document for project	20 days	02/02/24	21/02/24
2	Development	84 days	22/02/24	16/05/24
2.1	Sprint 1	21 days	22/02/24	13/03/24
2.2	Sprint 2	21 days	14/03/24	03/04/24
2.3	Sprint 3	21 days	04/04/24	24/04/24
2.4	Sprint 4	21 days	25/04/23	15/05/24
3	Project's Meeting	1 day	17/05/25	17/05/24
4	Final Release	1 day	17/05/24	17/05/24

8. Resource Allocation

Team's Information

Full Name	Phone	Email	Position
Thuan, Nguyen Tan Ph.D.	0905626276	thuannt.it.dtu@gmail.com	Mentor
Duy, Huynh Gia	0899249275	huynhduy25072002@gmail.com	Scrum Master
Minh, Huynh Tan Quang	0929097985	liverpool990minh12@gmail.com	Team Member
Dinh, Le Nguyen Nhut	0965925991	hoanphuoc20021208@gmail.com	Product Owner
Sy, Nguyen Van	0377445085	vansy01682937260@gmail.com	Team Member

9. Budget and Cost Estimate

Table 1: *Cost person/hours*

Full Name	Role	Salary Rate (USD/hour)
Duy, Huynh Gia	Scrum Master	2
Dinh, Le Nguyen Nhut	Product Owner	2
Sy, Nguyen Van	Team Member	2
Minh, Huynh Tan Quang	Team Member	2

Table 2: *Total cost estimate*

No.	Criteria	Price	Amount	Total (USD)
1	Working hours	\$2	2040	\$4080
2	Party	\$4	3	\$48
Total cost				\$4128

Description	Amount	Unit
-------------	--------	------

Number of members	4	Person
Number of working-hour per day	6	Hours
Number of working-day per week	5	Days
The duration of the project	120	Days
Party cost per member per time	2	USD

Explain:

- Amount of working hours = 4 members * 6 hours * 85 days = 2040 hours
- Amount of party cost support = 4 members * \$4/person * 3 times = \$48

10. Risk Management

Risk ID	Risk Description	Mitigation Plan	Contingency Plan
1	Data Quality and Availability	Conduct thorough data validation and preprocessing.	Explore options for augmenting datasets or collaborating with healthcare institutions for access to high-quality data.
2	Inaccurate cancer prediction	Use accurate and reliable medical data from validated sources	Provide clear and detailed information about the accuracy of predictions Guide users on the limitations of predictions and encourage them to seek advice from medical professionals
3	Legal and ethical issues	Strictly adhere to legal and ethical regulations in the healthcare field Consult and work with legal experts	Use clear and transparent contracts and service terms Update and enforce data protection and privacy measures
4	Technical Challenges	Establish a comprehensive testing framework.	Maintain clear communication channels to address technical issues promptly.

5	Resource Constraints	Conduct thorough resource planning and allocation.	Prioritize tasks based on criticality and explore options for outsourcing or partnering with external vendors.
6	Scope Creep	Establish clear project scope and objectives.	Implement change management processes and regularly review project progress against defined scope.
7	Information security	Implement strong security measures such as data encryption, authentication checks, and continuous monitoring	Respond promptly and effectively to any breaches or cyberattacks Regularly backup data to facilitate quick recovery in case of incidents

11. Stakeholder Engagement

Stakeholder	Role	Responsibilities	Level of Involvement
Project Sponsor	Executive	Provides project funding and strategic direction.	High, decision-making authority
Project Manager	Project Lead	Oversees project planning, execution, and delivery.	High, day-to-day management
Development Team	Technical Team	Develops and implements the cancer detection system.	High, hands-on development
Healthcare Professionals	End Users	Utilizes the system for cancer diagnosis and treatment planning.	Medium to High, user feedback and testing
Data Scientists	Subject Matter Experts	Provide expertise in machine learning and data analysis.	Medium to High, model development and validation
Regulatory Compliance Officer	Compliance Specialist	Ensures adherence to regulatory standards and guidelines.	Medium to High, regulatory compliance

Quality Assurance Team	QA Specialists	Conducts testing and quality assurance checks.	Medium, ensuring system reliability
Marketing and Sales Team	Marketing Specialists	Promotes the system to healthcare institutions and professionals.	Medium to Low, customer outreach
IT Support Team	Technical Support	Provides technical assistance and troubleshooting.	Low, as needed technical support

12. Project Constraints

Constraint	Constraints Description	Guidelines for Acceptance
Economic		Elements for consideration are design costs, production costs, maintenance costs, operating costs, and sales price
Environmental		The impact of the design on the environment as well as the impact of the environment (e.g., temperature range, humidity, vibration, electromagnetic interference immunity, and shock) on the design should be considered. Design for recycling and design to use recycled materials should also be considered
Ethical		Ethical considerations can be broad. Areas that are typically addressed include intellectual property, reverse- engineering,

		privacy, security, and the conflict between cost and safety
Public health, safety, and welfare		Includes safety standards as well as the impact of the design on users (for example, Electrical or physical hazards)
Social and Global		Addresses aspects such as benefits, risks, the man-machine interface, the acceptance of products by the intended user or by society at large, and global and social responsible engineering.
Database		The database of clinical signs, symptoms, genotypes, and genetic mutations is still limited and incomplete to make an accurate diagnosis in all cases.
Sustainability		Refers to the sustainability of resources, including material, energy, supplies, manufacturing techniques, personnel, operation, and the need for additional infrastructure, as well as the sustainability of the design including reliability, lifetime, durability, reusability, and maintainability.

13. References

No.	References	Document Information
1	Scrum Model	https://en.wikipedia.org/wiki/Scrum_(software_development)
		https://www.atlassian.com/agile/scrum
		https://www.digite.com/agile/scrum-methodology/
		https://docs.microsoft.com/en-us/azure/devops/boards/sprints/best-practices-scrum?view=azure-devops
		https://www.scrum.org/resources/scrum-guide
2	Technical	https://cloud.google.com/dialogflow/docs/
		https://slack.com/
		https://docs.spring.io/spring-framework/docs/3.2.x/spring-framework-reference/html/mvc.html
		https://developers.google.com/identity/sign-in/web/sign-in
		https://medium.com/
		https://reactnative.dev/docs/getting-started
3	Software Engineering Standards	https://www.nws.noaa.gov/oh/hrl/developers_docs/General_Software_Standards.pdf
		https://standards.ieee.org/standard/12208-2017.html
		https://sw-eng.larc.nasa.gov/

14. Attachment: DESCRIPTION OF PRODUCT REQUIREMENTS FORM**15. Approval**